since $x_p[n]$ and $x[n]$ are equal at integer multiples of N. The operation of ex
Nth sample is commonly referred to as *decimation.*[3] The relationship betwee
and $x_b[n]$ is illustrated in Figure 7.34.

To determine the effect in the frequency domain of decimation, we wis
the relationship between $X_b(e^{j\omega})$—the Fourier transform of $x_b[n]$—and X
end, we note that

$$X_b(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} x_b[k]e^{-j\omega k},$$

or, using eq. (7.48),

$$X_b(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} x_p[kN]e^{-j\omega k}.$$

If we let $n = kN$, or equivalently $k = n/N$, we can write

$$X_b(e^{j\omega}) = \sum_{\substack{n=\text{integer} \\ \text{multiple of } N}} x_p[n]e^{-j\omega n/N},$$

and since $x_p[n] = 0$ when n is not an integer multiple of N, we can also w

$$X_b(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x_p[n]e^{-j\omega n/N}.$$

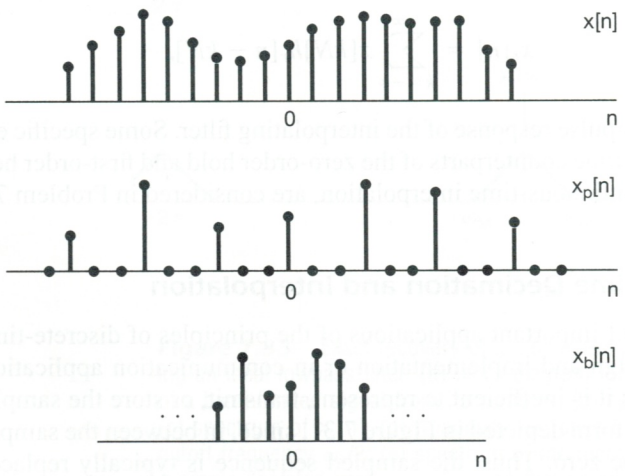


Figure 7.34 Relatio
$x_p[n]$ corresponding to
$x_b[n]$ corresponding to

[3]Technically, decimation would correspond to extracting every *tenth* sample. Howev
common terminology to refer to the operation as decimation even when N is not equal to 10.